Post-quantum cryptography on drone and robot computers: 2-page brief

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Contents

Bottom line. Verification-only post-quantum designs fit Cortex-M-class flight and robot computers; per-message post-quantum signatures do not fit MAVLink-class radio links; and the real deadline pressure is on **firmware-signature trust anchors that cannot be changed after shipping**. Everything below is published or derived from retrieved sources; **nothing was measured on a microcontroller** (hardware plan included).

What costs what (published; pqm4 on Cortex-M4F; board and clock unresolved (README says 24 MHz, default board code says 20 MHz); setups differ)

Operation	Cycles	Stack	Notes
ML-KEM-768 decaps (M4F code)	707,827	6,452 B	stack-optimised variant: 714,194 cycles in 2,844 B
ML-DSA-44 verify (M4F code)	1,421,623	8,912 B	2,712 B stack in the stack-optimised variant (3.2 M cycles)
ML-DSA-44 sign (mean, M4F code)	3,943,121	44,816 B	long tail from rejection sampling; 12.1 M cycles in 5,080 B stack variant
SPHINCS+ SHA2-128s sign (round-3, not final SLH-DSA)	7.66 billion	2,556 B	verify 7.47 M cycles

Sizes: ML-DSA-44 signature 2,420 B (65: 3,309; 87: 4,627); ML-KEM-768 key 1,184 B, ciphertext 1,088 B; Ed25519 signature 64 B; MAVLink 2 signature field **13 bytes**. One ML-DSA-44 signature is about 10 full MAVLink payloads (derived), about 0.34 s of a 64 kbps SiK radio (derived).

Headline findings

1. *Authenticity risk is prospective, secrecy risk is retroactive.* Move long-lived-secret links (video, site maps, customer data) to hybrid ML-KEM key exchange early; move **firmware trust anchors** early too, because a classical key frozen in boot ROM or fuses must outlive the device.
2. *Planning dates exist:* UK NCSC 2028 / 2031 / 2035; EU end-2026 / end-2030 / 2035 and an explicit line that products living beyond 2030 should have a post-quantum signature path for updates (Cyber Resilience Act link, Dec 2027); NIST IR 8547 draft: deprecate after 2030, disallow after 2035 (*in flux*). **No UAE algorithm-level or embedded-specific guidance**

found; the UAE National Encryption Policy (Nov 2025) requires government entities to plan transitions (text not retrieved).

3. *Upstream drone/robot software has no PQC yet:* none found in MAVLink, PX4/ArduPilot signing, DDS-Security 1.2, SROS2 or MCUboot. wolfBoot (ML-DSA, LMS/HSS, XMSS, hybrid), TF-M BL1 (LMS) and the PSA PQC API extension exist.
4. *Scheme choice for firmware:* LMS/HSS if one HSM-backed signer can guard state (smallest code, 1.4-2.6 kB signatures); ML-DSA if not; SLH-DSA if conservative assumptions outweigh 7.9 kB+ signatures. Ship **dual** (classical + PQ) signatures, A/B slots, anti-rollback.
5. *Fleet paths:* firmware update suffices only if there is an updatable stage before application code with flash/RAM headroom; classical-only immutable ROM roots need replacement, a gateway, or a documented tolerated risk.

Migration in five phases: inventory (now) -> crypto-agility groundwork -> PQ-capable verifier and dual signing in the updatable bootloader -> hybrid session keys and links -> procurement requirements and classical retirement. Each phase has exit criteria, risks and rollback in `docs/MIGRATION_PLAYBOOK.md`.

What remains unmeasured: any Cortex-M33 result; final-parameter LMS and SLH-DSA on any MCU; measured energy for the final standards; flight-controller (Cortex-M7) numbers; the harness itself on hardware. Verification: 0 wrong values in 48 verifiable sampled rows (about 6% upper bound).

First experiment (E1): ML-DSA-44 verify and ML-KEM-768 decaps cycles, stack and flash on a NUCLEO-U575ZI-Q (M33) and a NUCLEO-L4R5ZI (M4F), pinned mlkem-native/mldsa-native v2.0.0, 3 power cycles, medians with bootstrap CIs (about $60 of boards).